

PART A

Constraints:

- ➔ It is a rule applied in a table to ensure accurate and consistent data to be stored in database

Primary Key:

- ➔ Uniquely identify each record in a table. The primary key should not be duplicate values and NULL values

Foreign Key:

- ➔ Foreign creates a relationship between tables by identifying the primary key of the table

Candidate Key:

- ➔ Combination of column that can uniquely identify the row

Alternate key:

- ➔ A Candidate key which is not chosen as a primary key is an Alternate key

Composite key:

- ➔ A Composite key is a two or more Columns that uniquely identify the records

Super key:

- ➔ Super Key is a set of one or more columns that uniquely identifies the records

NOT NULL:

- ➔ The NOT NULL constraint ensures that a column cannot contain NULL values.

CHECK:

- ➔ The CHECK constraint ensures that the values satisfy a specified condition.

DEFAULT:

- ➔ The DEFAULT constraint automatically assigns a default value if no value is provided

Why are constraints required?

- ➔ Constraints are required to maintain data accuracy, consistency, in a database.

PART B

1. Compare primary key and Unique key

Primary Key	Unique Key
Uniquely identifies each record	Ensures unique values
Cannot contain NULL values	Can contain one NULL value
Only one Primary Key in a table	Multiple Unique key can exist

2. Primary Key vs foreign Key

Primary Key	Foreign key
Uniquely identifies a record	Links one table to another
Links one table to another	Can contain duplicate and NULL value
Only one primary key	Multiple Foreign Keys allowed

3. Candidate key vs super key

Candidate key	Super key
Uniquely identify the record with the minimum columns	Uniquely identify the records may contain extra column
Every candidate is super key	Not every super key is a candidate key

4.Candidate Key vs Alternate Key

Candidate key	Alternate key
All possible unique columns	Candidate key which is not choose as a primary key
Include the primary key	Does not include primary key

5.Composit key vs Primary key

Composite key	Primary key
User two or more columns	Use one column in per table
Combinations identifies of record	Main key of the table

6. Constraints vs Keys

Constraints	Keys
Role applied to columns	Used to identify records
Maintain data integrity	Maintain uniqueness and relationship

7. NOT NULL vs CHECK

NOT NULL	CHECK
Do not contain null value	Restricts value based on condition

8. UNIQUE vs PRIMARY KEY

UNIQUE	PRIMARY KEY
Prevent duplicate values	Prevents Duplicate and null values
Multiple unique constraints allowed	Only one primary key allowed

PART C – Identify Keys

1. Primary keys

Department_ID (Department table)

Employee_ID (Employee table)

Project_ID (project table)

2. List all candidate keys

Department_ID (Department)

Employee_ID, Email, PAN_Number (Employee)

Project_ID (Project)

3. List all alternate keys

Email, PAN_Number (Employee)

4. Five super keys

Department_ID

Employee_ID

Email
PAN_Number
Project_ID

5. Which Column Should be Unique

Email, PAN_Number

6. Which Column should NOT NULL

Employee_Name, Email, PAN_NUMBER, Department_ID Project_Name

PART D – Interview

1. Can a table have multiple primary key?

No , A table must contain only one primary key

2. Can you table have multiple candidate keys?

Yes ,A table can have multiple candidate keys

3. Can a foreign tree contain duplicate values?

Yes ,a foreign key can contain duplicate values

4. Can Unique contain NULL?

Yes, Unique can contain one null value

5. Can composite key have more than two columns?

Yes, Composite key can have more than two columns

6. Is Every candidate key is a super key?

Yes, Every candidate is a super key

7. Is Every super key is a candidate key?

No ,super key may contain extra colmn whereas candidate key is the minimal key.

8. What is referential integrity?

Referential integrity ensure that foreign key values always refer to valid primary key values

9. Which Constraints maintain referential integrity?

FOREIGN KEY constraints